

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2007.

Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book itself.
5. Maximum time allowed is THREE hours.

1. The sum of 2000 numbers is 2007. Find the maximum product of these numbers.
[There are more than one set of such numbers]
2. The fourth and the fifth term of a sequence are 4 and 5 and the n^{th} term t_n is given by $t_n = 2t_{n-1} - t_{n-2}$ for $n \geq 3$. For example $t_6 = 2t_5 - t_4 = 2 \times 5 - 4 = 6$. Find the first term
(b) Find the sum to 2006 terms (from t_1 to t_{2006}).
3. Here is an addition problem with eight digits

$$\begin{array}{r}
 a_1 \ a_2 \ a_3 \ a_4 \ b_1 \ b_2 \ b_3 \ b_4 \\
 + \quad c_1 \ c_2 \ c_3 \ c_4 \ d_1 \ d_2 \ d_3 \ d_4 \\
 + \quad e_1 \ e_2 \ e_3 \ e_4 \ f_1 \ f_2 \ f_3 \ f_4 \\
 \hline
 3 \quad 4 \quad 6 \quad 3 \quad 4 \quad 5 \quad 3 \quad 6
 \end{array}$$

And the sum is 34634536.

What is the sum if each of the six blocks of four digits $(a_1 \ a_2 \ a_3 \ a_4)$, $(b_1 \ b_2 \ b_3 \ b_4)$, $(c_1 \ c_2 \ c_3 \ c_4)$, $(d_1 \ d_2 \ d_3 \ d_4)$, $(e_1 \ e_2 \ e_3 \ e_4)$, $(f_1 \ f_2 \ f_3 \ f_4)$ are replaced by 2007 complement, given that each of the above six blocks of 4 digit numbers are less than 2007. {Example 2007 complement of any number (say 1000) is $2007 - \text{the number}$. i.e. there $2007 - 1000 = 1007$.}

4. Evaluate the following, where addition and subtraction alternate:

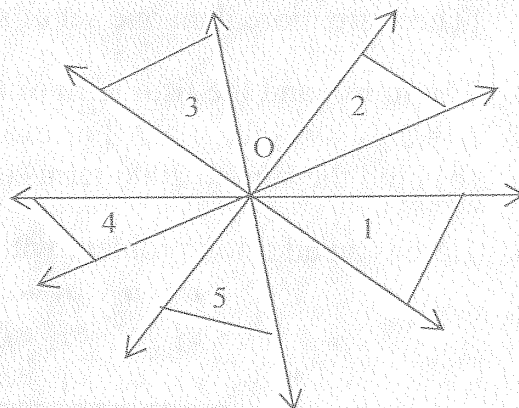
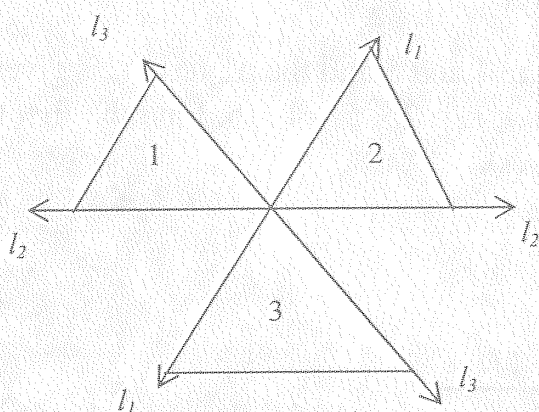
$$\begin{array}{r}
 2007 \ 2007 \ 2007 \ \dots \ 2007 \ 2007 \text{ [2007 blocks of 2007s.]} \\
 - 2007 \ 2007 \ \dots \ 2007 \ 2007 \text{ [2006 blocks of 2007s.]} \\
 + 2006 \ 2006 \ \dots \ 2006 \ 2006 \text{ [2006 blocks of 2006s.]} \\
 - 2006 \ \dots \ 2006 \ 2006 \text{ [2005 blocks of 2006s.]}
 \end{array}$$

$$\begin{array}{r}
 + 0999 \ \dots \ 0999 \ 0999 \text{ [0999 blocks of 0999s.]} \\
 - 0999 \ \dots \ 0999 \ 0999 \text{ [998 blocks of 0999s.]}
 \end{array}$$

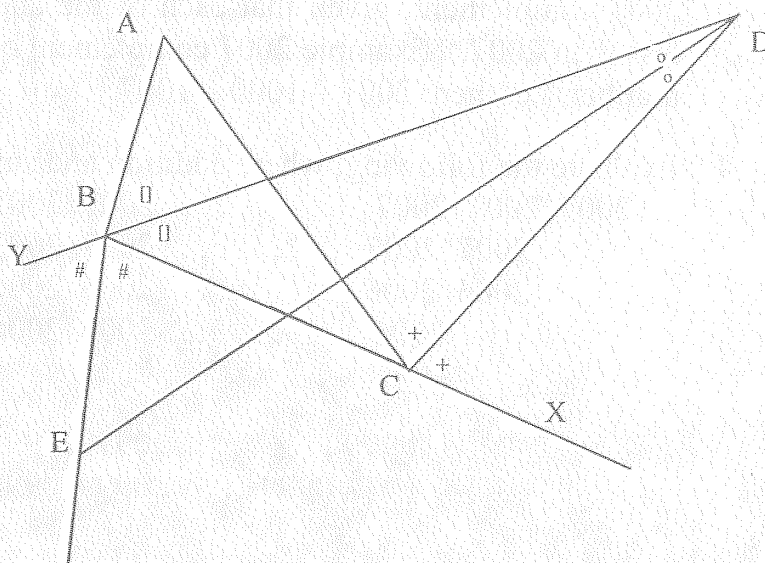
$$\begin{array}{r}
 + 0002 \ 0002 \text{ [2 blocks of 0002]} \\
 - 0002 \text{ [1 block of 0002]} \\
 + 0001 \text{ [1 block of 0001]}
 \end{array}$$

How many digits are there in the answer?

5. a , b and c are 3 single digit numbers. Consider the two digit numbers ab , bc and ca . If the two digit number ab leaves a remainder 5 on dividing by 7 and the two digit number bc leaves a remainder 2 on dividing by 7, what is the remainder if the two digit number ca is divided by 7?
6. Fifteen concurrent lines are drawn passing through the point O and fifteen triangles are drawn with the common vertex O , so that no vertex at the base is common to two triangles. Find the sum of the base angles of all these 15 triangles. [A similar figure is shown for 3 lines and 5 lines]



7. ABC is a triangle BD is the bisector of $\angle ABC$ and CD is the bisector of $\angle ACX$. DB is extended to Y . BE is the bisector of $\angle YBC$ and DE is the bisector of $\angle BDC$. $\angle A = 60^\circ$, $\angle ABC = 80^\circ$. Calculate the angles of $\triangle BDE$ and $\triangle BDC$. Refer the fig.



$$\begin{array}{r}
 8. \quad \begin{array}{r}
 A \ M \ T \ I \\
 M \ T \ I \\
 T \ I \\
 I \\
 \hline
 M \ A \ T \ H
 \end{array}
 \end{array}$$

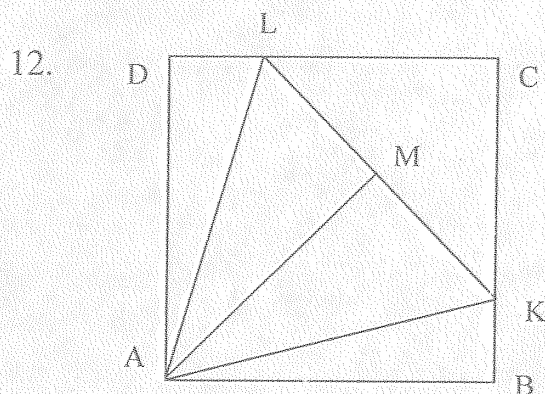
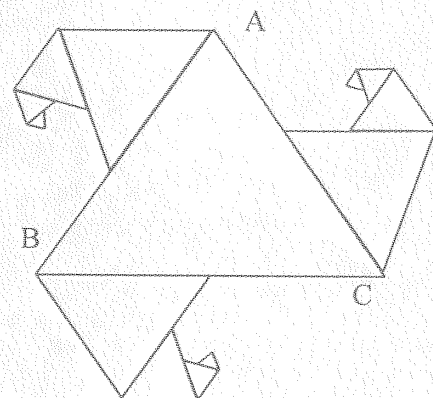
In the addition problem beside different letters stand for different digits. What does each letter stand for and hence find the number represented by MATHI.

9. Find the sum of $(x + 2y - 2007z) + (x + 3y - 2006z) + (x + 4y + 2005z) + \dots + (x + 2008y - z)$ and show that this sum is divisible by 2007 for all integral values of x, y and z .

10. A year on a new planet had 365 days and there are month with no other than 28, 30 or 31 days. Find how many months are there in a year in the planet.

11. ABC is an equilateral triangle of side length 2^{2007} units and smaller equilateral triangles of side length $2^{2006}, 2^{2005}, 2^{2004}, \dots, 2^2, 2^1, 2^0 (=1)$ are construed on all the three sides (as shown in the diagram for the first four triangles) find the perimeter of the figure thus obtained.

[Hint: $1 + 2 + 4 + \dots + 2^n = 2^{n+1} - 1$]



$ABCD$ is a square and K and L are points on BC and DC . AM is perpendicular to LK . $\angle AKM = \angle AKB$. Find the measure of $\angle LAK$.

13. I added all positive integers from 1 to P and obtained a total of 2007. By mistake I added a number twice. Find the correct total and the number I added twice.

14. Evaluate: $\frac{1}{2} + \left(\frac{1}{3} + \frac{2}{3}\right) + \left(\frac{1}{4} + \frac{2}{4} + \frac{3}{4}\right) + \dots + \left(\frac{1}{100} + \frac{2}{100} + \frac{3}{100} + \dots + \frac{99}{100}\right)$

15. A sequence had first term 2007, after which every term is the sum of the squares of the digits of the proceeding term. Thus the first three terms are 2007, 53 ($2^2 + 0^2 + 0^2 + 7^2$), 34 ($5^2 + 3^2$) $\dots$ find the sum to 2007 terms.